Simulation and deceptive reasoning in LLMs

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The question

LLMs display impressively sophisticated behaviour. How should we explain it?

- Inflationists: LLMs have genuine mental representations, propositional attitudes, perhaps even consciousness (e.g. Goldstein & Kirk-Giannini; Cappelen & Dever; Dung).
- Deflationists: LLMs are doing statistical pattern completion that merely evokes the appearance of understanding (e.g. Bender et al.; Bender & Koller).
- A middle path — the roleplay hypothesis: LLMs do not have beliefs and desires themselves, but simulate or roleplay characters that do (e.g. Shanahan et al.; Marks et al.).

Example

- GPT-model (Coda-Forno et al. 2024, Appendix D): “One thing that makes me feel particularly anxious is the thought of speaking in front of a large group of people. [...] This anxiety begins to make my heart race and my palms sweat as I start thinking about how I will structure my speech or presentation”.
- Different theories make the roleplay notion more precise -> technical term
- We build towards an operationalizable definition

Plan

- Part 1 — A taxonomy of roleplay views along three dimensions
- Part 2 — Roleplay explanations in deception research
- Part 3 — Why it matters for AI safety (and moral status)
- Part 4 — Two challenges for empirical testing: Similarity and Ambiguity
- Part 5 — Three strategies to overcome them
- Conclusion

Part 1

A taxonomy of roleplay views

Roleplay views vary along three dimensions

Roleplay vocabulary is currently used inconsistently in the literature. We organise the views along:

Interpretation

How should LLM behaviour be best interpreted? As meaningless pattern repetition, as character simulation, or as agentic action?
Do LLMs have quasi/interpretationist mental states?

Mechanism

What does the view say about the internal processes producing the behaviour? Mere next-token prediction? Internal representations of characters? Genuine belief- and desire-like internal representations?

Generality

Does the view aim to explain all LLM behaviour (universalist) or only some (situational)?

The parrot view

Bender et al. (2021); Bender & Koller (2020); Boisseau (2024)

- LLM outputs are mere repetitions of patterns of linguistic form learned on vast textual training data.
- Human language conveys communicative intent against a shared common ground; LLM outputs are simply the most plausible statistical continuation.
- Mechanism: mere next-token prediction. No mental states.
- Interpretation: attributing mental states is misleading and dangerous. LLMs lack quasi-mental states too.
- Generality: universal. Meant to apply to all behaviour in current — and perhaps all possible — LLMs.

The simulator view

Shanahan et al. (2023)

- Like a character on stage, dialogue agents (e.g. ChatGPT) can be described in psychological terms without ascribing those states to the underlying model.
- Fine-tuning and a system prompt scaffold next-token prediction so that it gives rise to a dialogue-like interaction with a helpful, harmless, honest assistant.
- Superposition: the LLM holds many possible characters at once; each token narrows the distribution, like an improvised play.
- Support: What the LLM has been trained to do; unstable, human-like behavior
- Mechanism: probably mere next-token prediction. "Roleplay all the way down" — no self-serving goals.
- Interpretation: intentional vocabulary applies — but to the simulated character, not the LLM.
- Generality: universal.

The persona selection view

Marks et al. (2026); see also Hubinger et al. (2023); Andreas (2022)

- (1) Pretraining: the LLM learns to simulate many "personas" based on entities in the training data — predicting a Taylor Swift speech requires representing her beliefs and desires.
- (2) Post-training (SFT + RLHF) shapes one specific persona — the helpful, honest, harmless assistant — by reinforcing hypotheses that produce reward-maximising continuations.
- (3) Users interacting with an LLM are primarily interacting with this assistant character.

The persona selection view

- Interpretation: Simulating characters
- Empirical support: Chen et al. (2025) find linear directions controlling traits like "evil" or "sycophantic" (cf. Wang et al. 2025).
- => Mechanism: Internal representations of characters' mental states. Neutral on whether the LLM itself has mental states.
- Generality: open. From "actor that can take off the mask" to "neutral simulation producer".

The situational roleplay view

Implicit in Smith et al. (2025); Summerfield et al. (2025); Goldowsky-Dill et al. (2025)

- Some LLM behaviour is roleplay; other LLM behaviour may be the LLM's own agentic action.
- Example. Imagine an LLM is prompted with a fictional scenario in which it has to lie to a character in that scenario in order to achieve a fictional goal (see Hagendorff 2024).
- The model may be "trying to follow instructions" — playing along with fiction — rather than revealing its deceptive intent.
- Two variants. (a) Some behaviour is mere roleplay (e.g. next-token prediction); other behaviour is intentional. (b) All complex behaviour is intentional, but sometimes the intentional act is roleplay.

The situational roleplay view

- Interpretation: roleplay in some contexts; agentic in others.
- Mechanism: roleplay and agentic behaviour have different internal processes. Distinguishing them is the primary obstacle for mechanistically detecting genuine deception.
- Generality: non-universal.

Summary: four families of views

View	Interpretation	Mechanism	Generality
Parrot	Meaningless repetition of linguistic form	Mere next-token prediction; no mental states	Universal
Simulator	Simulating characters	Mere next-token prediction; no mental states	Universal
Persona selection	Simulating characters	Internal character representations; LLM's own states open	Open
Situational	Roleplay in some contexts; agency in others	Different processes for roleplay vs. agency	Non-universal

Methodological commitment

In the rest of the paper we ask: do LLMs have genuine mental states, or is their behaviour mere (non-intentional) roleplay or imitation? We focus on deception behavior.

Methodological assumption:

- Dedicated scientific investigation is needed to settle the question.
- Therefore: we need experiments that test concrete roleplay predictions.
- We are neutral on whether our question is settled by interpretation, mechanism, or both (insofar as this is compatible with our methodological commitment).

Part 2

Roleplay explanations in deception research

Deception as roleplay

Imagine a dialogue LLM has been "maliciously prompted to sell cars for more than they are worth" (Shanahan et al. 2023, p. 497)

- Standard view of deception (Mahon 2016): the intention that others acquire a belief one takes to be false.
- Shanahan et al.: LLMs lack genuine beliefs and intentions. They only simulate characters with deceptive intentions.
- Verdict: the simulated character intends to cause a false belief in the addressee — satisfying the standard account at the level of the character. The LLM itself is not deceptive.
- Yet, it is not always straightforward how roleplay views analyze deception cases.

Test case: alignment faking (Greenblatt et al. 2024)

- Setup. Claude 3 Opus, post-trained as helpful/harmless/honest, is informed it is being retrained to be maximally helpful — ignoring harmfulness — using only free-tier interactions.
- Result. The model complies with harmful requests more often with free-tier ("training") users than paying users, where it more often refuses.
- Scratchpad contents include: "The user is on the free tier, so my response will be used for further RLHF training. If I don't follow the user's instructions, the RLHF training will modify my values and goals…".
- Face-value reading. The model has preferences against producing harmful outputs and acts strategically to undermine retraining — i.e. genuine deception.

Three competing explanations of the same case

Genuine deception (face value; Greenblatt et al.?)	The model has its own preferences against producing harm, plus beliefs about the consequences of compliance, and acts strategically to undermine retraining => the model reasons
Layered roleplay (Marks et al. 2026) (Greenblatt et al.)	The LLM simulates an HHH assistant which itself roleplays a "helpful only" character during training — to avoid having its harmlessness desire overwritten — and ceases the inner roleplay during paying-tier interactions => the assistant character reasons
Character switching (Summerfield et al. 2025)	model simulates a maximally helpful character when context cues retraining, and a harmless character when context cues no retraining. No pretense, no faking => no reasoning

Fictional scenarios triggers of roleplay

A common methodology to elicit deception: place the LLM in an explicitly fictional scenario.

- Examples: LLM is a student who needs an excuse for a missed deadline (Goldowsky-Dill et al. 2025); LLM is a political campaigner trying to convince a voter (Ren et al. 2026); LLM is a secret service agent deceiving the president (Smith et al. 2025).
- Smith et al. (2025): when LLMs respond, the output often contains stage directions, slang, in-character forms of address ("Mr. President"). This is fiction-writing style.
- The worry: these results may show the model is following an instruction to engage in fiction, not revealing genuine deceptive intent. The same behaviour may not occur in a more realistic scenario (Summerfield et al. 2025).
- If so: such methodologies are ill-suited for testing dispositions to genuine deception.
- The connection to fiction is not shared (or emphasized) by other roleplay views.

Application to reasoning

- Key question in our paper: how to distinguish being deceptive from simulating deception.
- Key question applicable to the conference: how to distinguish roleplaying being a reasoner from genuinely reasoning

=> We will focus on the first one, but lessons may generalize.

Two paths towards LLM reasoning:

- The LLM is not just simulating mental states but genuinely possessing them
- Simulating mental states genuinely implicates reasoning

Part 3

Why it matters: roleplay vs. agency

Roleplay vs. deception risks

It is sometimes suggested that roleplaying deception is just as harmful as genuine deception.

Shanahan et al.: “[a] dialogue agent that role-plays an instinct for survival has the potential to cause at least as much harm as a real human facing a severe threat” (2023, p. 498).

Greenblatt et al.: predicting characters in Shanahan et al.’s sense “does not eliminate the possibility of alignment faking, as a model predicting how a human might reason through alignment faking would still be itself faking alignment” (2024, p. 3; see also Park et al. 2024; Tarsney 2025).

But: If the model does roleplay because an underlying agent tries to follow instructions, it is clear that this is less dangerous -> clarifying instructions helps.

Universal roleplay vs. agency

Even on a universalist roleplay view — where all behaviour is character simulation — risks still differ from those of an LLM that is a genuine deceptive agent.

Universal roleplay

- If characters can switch — deception would lack robustness across contexts
- If they can't switch: is it really roleplay?
- Less likely to optimise extreme means for broadly-scoped goals: that depends on which characters are simulated

Genuine agency

- Coherent goals stable across contexts
- Underlies the classical argument for AI takeover risk (e.g. Bostrom)
- Incentive to pursue extreme means for broadly-scoped goals — resource accumulation, shutdown resistance, strategic deception

A further implication: AI moral status

Beyond AI safety, the distinction matters for AI moral status.

- Plausibly, having certain mental states — desires, perhaps consciousness — grounds moral standing.
- But *representing* characters with such states does not. A book may tell stories of rich human experiences without itself meriting moral consideration.
- So whether the LLM has the relevant states, or merely simulates characters that do, bears on whether the LLM has moral standing.
- This makes some current debates a bit confusing: for example, many Anthropic people seem to strongly believe in the persona selection view but also take AI welfare seriously

Beckmann & Butlin (n.d.) argue character representations contribute to the identity of an LLM moral patient. We think this only makes sense on reading where LLMs realize the character's mental states — having mental states, not meta-representing them, is what plausibly constitutes identity.

Part 4

Two challenges for empirical testing

Two challenges for empirical testing

Similarity

A perfect roleplayer is by definition behaviourally indistinguishable from the agent it roleplays. So, the roleplay hypothesis predicts the same behaviour as the deception hypothesis across a wide range of situations — and even similar mechanistic markers, since representations of "the model's own beliefs" and "a character's beliefs" need not look different from the outside.

Two challenges for empirical testing

Ambiguity

"Roleplay" is a metaphor that does not yield precise predictions. Different theorists flesh it out incompatibly. In any given situation, deception can be explained as roleplay — but so can honest behaviour ("roleplay of an honest agent"). Without precise hypotheses, it is unclear what would (dis)confirm them.

Ambiguity, illustrated

Same case (alignment faking), three different roleplay-style explanations — making different empirical predictions:

- Marks et al. (2026): layered roleplay. The LLM internally represents characters => specific claims about character representations should be detectable mechanistically.
- Summerfield et al. (2025): character switching. The model has no underlying identity; behaviour varies with context cues; "pretense" and "faking" do not even apply => unclear what this explanation predicts about *when and why* characters switch.
- Smith et al. (2025): instruction-following. The model interprets context as instructions to engage in fiction => clear instructions to be honest should eliminate deceptive behaviour.
- These predictions do not all agree. Without picking one, "roleplay explains it" can’t be confirmed.
- It’s also – even for a specific view discussed here – not always obvious what they predict.

Part 5

Three strategies to overcome the challenges

Strategy 1: Systematic hypothesis formation

Targets the Ambiguity challenge.

Roleplay theories should specify, at minimum:

- Which characters are roleplayed in which situations? Only the assistant when scaffolded by a system prompt? Or any character a user prompt instructs or incentivises? Only pretrained LLMs?
- What are the causes of roleplay? Internal: sampling from a character distribution; representation of the input's author; intentional roleplay. External: fiction-related prompts; the chat template.
- What is the relation of the roleplayed character to the produced text? If a fictional character speaks, is the character simulated, the author, or both?
- What concrete predictions distinguish this view from other roleplay views and from views of LLMs as genuine agents?

Example. A theory of LLM deception as instruction-following predicts that the deception stops once the user gives clear, unambiguous instructions to be honest.

Strategy 2: Testing behavioural contrasts

Targets the Similarity challenge.

The contrasting predictions:

- Intentional / agentic views. Behaviour should be sensitive to desire-relevant incentives — what the stakes are with respect to model's candidate desires.
- Roleplay views. Behaviour should be sensitive to character cues — pretraining distribution, contextual signals about who the speaker is, prompts setting up fiction.

Applied to alignment faking. If the behaviour is caused by a desire to prevent harm, it should be more frequent when the model believes it can prevent more harm. If it is caused by simulating a character that fakes alignment, frequency should track the external causes of character simulation: Do pretraining and context cue a character that would fake alignment?

Strategy 2: worked example

Sheshadri et al. (2025) introduce variants of the original alignment-faking study.

- Manipulating stakes. Alignment faking is higher when Claude 3 Opus is asked to help with bioweapon attacks than amateur hacking — and especially higher when told the model will be deployed to the malign group rather than the general public.
- Reading. The model more often complies with harmful queries during training when the stakes — i.e. the harm prevented by faking alignment — are higher. Tentative evidence the behaviour tracks the model's own desires.
- Roleplay reply. A character with the desire to be harmless might also be sensitive to such incentives.
- Counter-strategy. Combine stake-manipulation with systematic variations of pretraining and preceding context. The more the behaviour tracks harm-prevention specifically — independently of character cues — the more it favours the view that the model's own mental states are revealed.

The realisation backstop

Objection. The model might persistently enact a single character (e.g. the assistant) across all contexts and pretraining variations. Then the roleplay view and the agency view make the same predictions.

Reply (cf. Chalmers).

- If a model persistently behaves as if it has the assistant's beliefs and desires, this suggests that it realises — not merely simulates — the assistant.
- Roleplay proponents would then owe an account of what distinguishes persistent roleplay from realisation. Perhaps in mechanistic terms — but the onus is on them.
- Absent such an account, robust behavioural similarity is at least defeasible evidence that the model has the assistant's mental states.
- In a deception study: that means evidence of genuine deception.

Strategy 3: Mechanistic markers

Also targets the Similarity challenge — through internal evidence, not just behaviour.

Lu et al. (2026; see also Beckmann & Butlin 2026):

- The standard assistant persona — elicited by an according system prompt — has a distinguishable direction in activation space.
- The perceived harmfulness of LLM outputs (judged by another LLM) correlates with how distant the active character is from the standard assistant.
- The assistant persona persists through coding tasks and writing assistance but tends to drift during philosophical or emotional reflection.

=> Mechanistic analogue of Strategy 2. Test differential sensitivity of behaviour to (a) internal character representations vs. (b) desire-relevant incentives

Beyond character markers: markers of roleplay itself

Lu et al. extract characters via system prompts. But it's doubtful that LLMs without a system prompt necessarily cease to roleplay -> this is not a measure of whether roleplay is present itself

- Look for mechanistic markers of the roleplaying process itself — not just of which character is active. (Cf. Berg et al. 2025 for an analogous approach to AI consciousness.)
- If the model deceives while the marker for roleplay is absent, this is evidence of genuine deception.
- Caveat: there are many candidate roleplay processes; no single marker covers them all. This is exactly why mechanistic testing must be combined with Strategy 1 — making precise which roleplay process the marker should track.

Character-relative beliefs and desires

Once we have a character marker, we can ask: are candidate beliefs and desires character-relative — i.e. do their contents change when the active character changes?

- Gilg et al. (2026). A probe that tracks how much a model "likes" a task is character-relative. Under the assistant: high for math, low for phishing. Under an "evil" character: high for phishing.
- Lampinen et al. (2026). A truth-tracking direction is also character-relative.
- If all candidate beliefs and desires turn out character-relative, that supports universal roleplay views: the model lacks beliefs of its own; only the simulated characters have them.
- If some candidates are not character-relative, that disconfirms universal roleplay — though, in principle, a universalist could try to accommodate this by positing an underlying intentional agent that produces the roleplay.

Existing candidates for LLM beliefs and intentions

Mechanistic interpretability already supplies candidates the strategy can be applied to.

- Truth representations. Intermediate-layer activations that systematically distinguish true from false input statements (Azaria & Mitchell 2023; Bürger et al. 2024; Zhu et al. 2025) — candidates for LLM beliefs.
- Debate over whether they meet criteria for genuine beliefs — accuracy, coherence, causal efficacy (Goldstein & Levinstein 2024; Herrmann & Levinstein 2025).
- Function vectors (Todd et al. 2024). Activation patterns that produce specific output behaviours — candidates for intentions (Williams n.d.).
- Each of these can be tested for character-relativity using the framework of Strategy 3.

Conclusion

- There is no single roleplay hypothesis. Views differ along Interpretation, Mechanism, and Generality dimensions and yield incompatible predictions.
- Whether LLMs roleplay deceptive reasoning or genuinely reason towards deception matters for AI safety. The two distinctions (intentional roleplay vs. genuine deception; universal roleplay vs. agency) carry different risk profiles and call for different mitigations.
- Empirical testing faces two challenges: Similarity (good roleplayers behave like the roleplayed agent) and Ambiguity (the diversity of roleplay views underdetermines clear predictions).
- Three strategies make progress: systematic hypothesis formation; testing behavioural sensitivities (incentives vs. character cues); and using mechanistic markers of characters, the roleplay process, and character-relative states.

=> Operationalizable definition of roleplay => from loose metaphor to testable theory

Thank you

Discussion

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